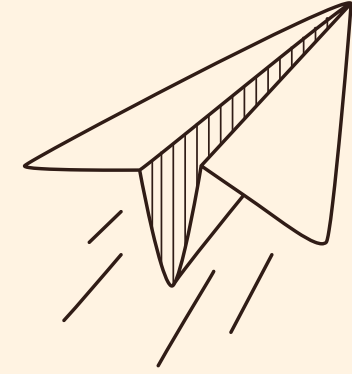
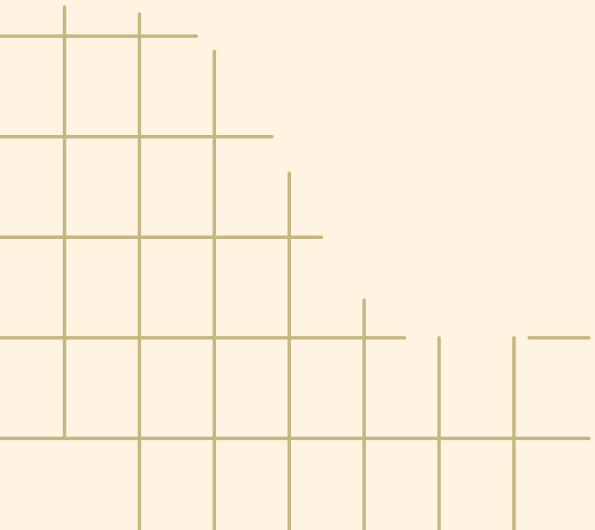




PRINCIPLES AND METHODS OF TEACHING

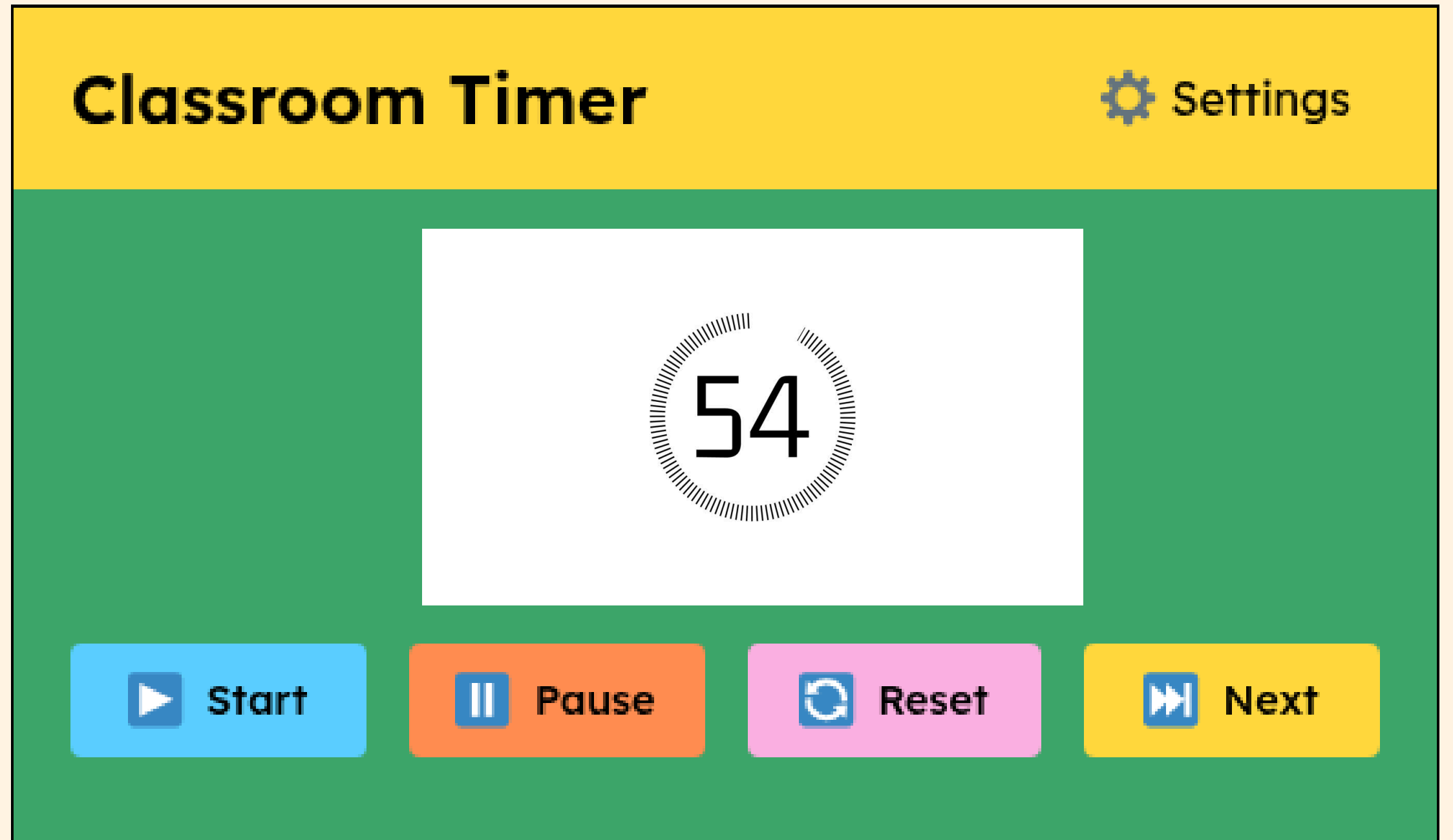
Unit 3: Understanding Nature and Principles of Teaching for Effective Instructional Planning

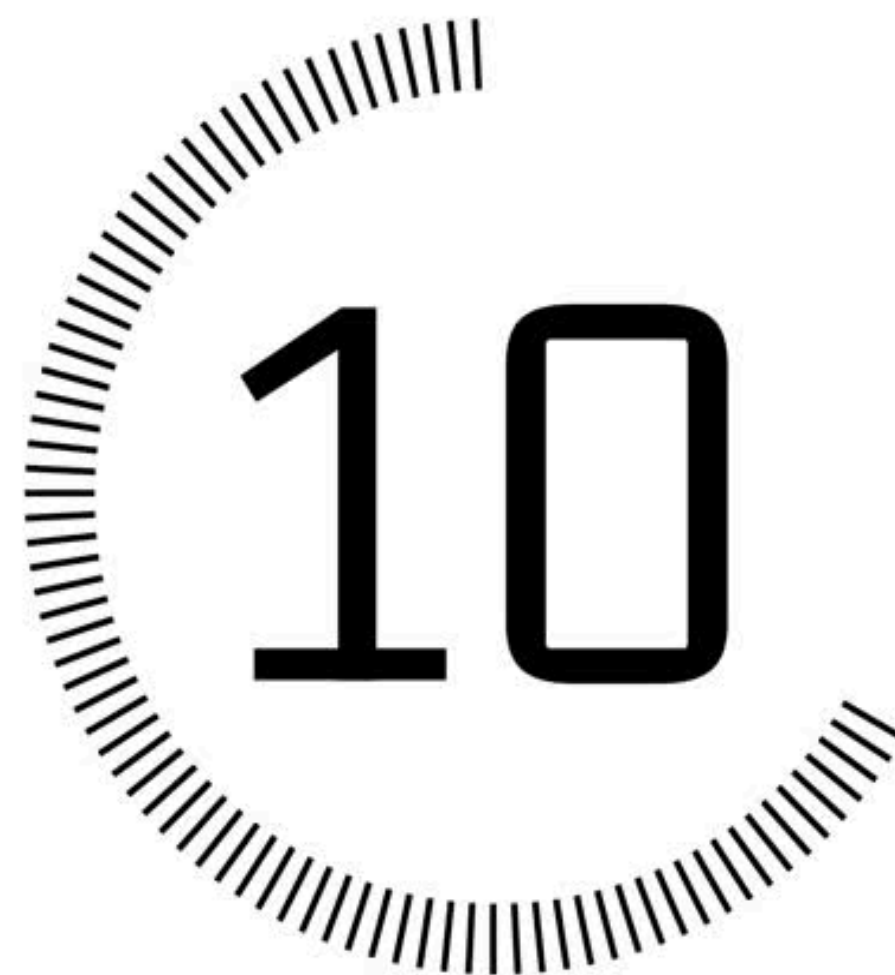


The Zoom Background Showdown

Imagine you have just been given a million dollars, but there is a catch: you are legally required to spend the next 6 months living in a fictional world, a historical era, or a specific dream vacation spot.

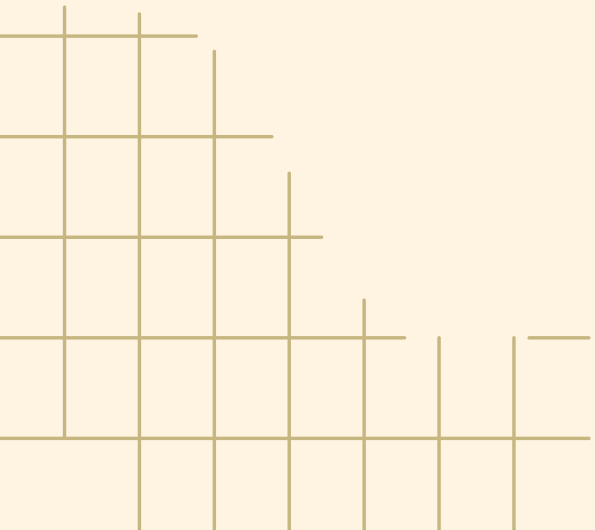
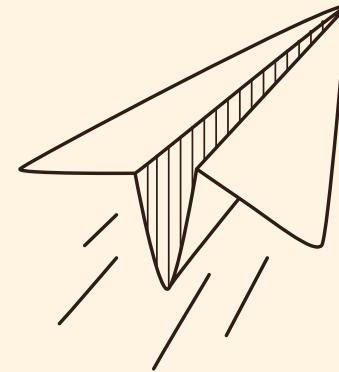
You have exactly *60 seconds* to go to Google Images, find a picture of that place, download it, and upload it as your Zoom virtual background.

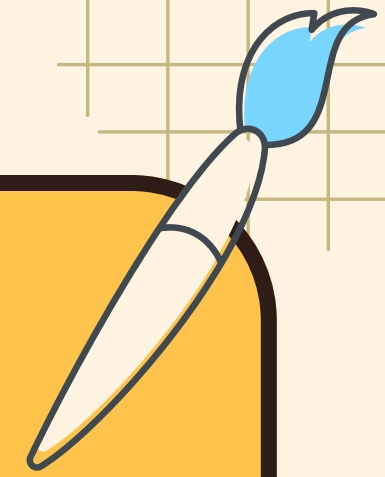






Universal Design for Learning



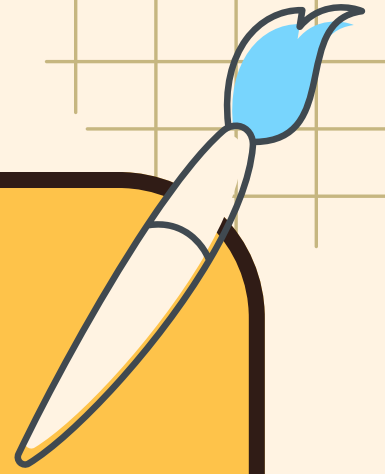


Universal Design for Learning

“UDL is a framework to improve and optimize teaching and learning based on what we know about the human brain. Each brain is made of billions of interconnected neurons that form unique pathways. Like fingerprints, no two brains are alike.”

- Center for Applied Special Technology (CAST)





**“If a plant does not grow, you
fix the soil and water, not the
plant.”**



Historical Background

CAST maps our learning processes into three primary brain networks (affective, recognition and strategic)

Mapping the Learning Brain (1990s)

Architect Ronald Mace, a wheelchair user, coins the term "Universal Design" (UD).

The Architectural Blueprint (1970s)

Global Inclusion & The Equity Shift (Present / UDL 3.0)

The Policy Integration: UDL expands globally. In the United States, it is written into federal education laws (ESSA). In the Philippines, UDL principles are adopted to guide DepEd's inclusive education policies and the localization of lesson plans.

The Radical Realization (1984)

Dr. David Rose (neuropsychologist) and Dr. Anne Meyer (educational psychologist) found CAST in Boston, originally trying to use computer technology to help students with learning disabilities.

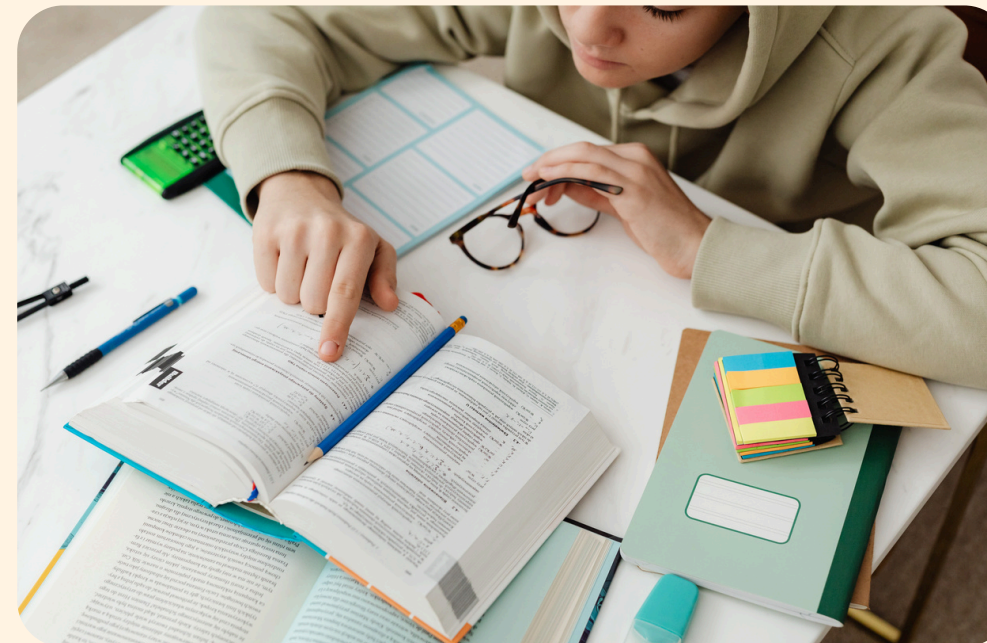
THREE CORE PILLARS



MULTIPLE MEANS OF ENGAGEMENT

“the *WHY* of learning”

Tap into learners’ interests, offer appropriate challenges and increase motivation



MULTIPLE MEANS OF REPRESENTATION

“the *WHAT* of learning”

Give learners various ways of acquiring information and knowledge



MULTIPLE MEANS OF ACTION & EXPRESSION

“the *HOW* of learning”

Provide learners the alternatives for demonstrating what they know

Universal Design for Learning Guidelines

The goal of UDL is [learner agency](#) that is purposeful & reflective, resourceful & authentic, strategic & action-oriented.



Grade 10 English

UDL SCANNING DIMENSION	THE MONOCULTURE (CURRENT DESIGN)	PREDICTED BARRIERS	THE UDL TOOL SOLUTION
Scan the Representation	The teacher plays a 20-minute audio recording of British actors reading Macbeth in Old English while students follow along in a dense printed textbook.	Students like Elena (ELL/Visual learners): They get completely lost in the rapid British accents and archaic words. They disconnect due to language overwhelm.	
Scan the Action & Expression	The summative performance task requires every student to individually write a formal 3-page character analysis essay in English	Students like Marcus (Reluctant writers/Dysgraphia): They know who Macbeth is and understand his ambition perfectly, but the physical act of essay layout blocks them, resulting in blank papers.	
Scan the Engagement	The classroom dynamic is entirely passive: silent, individual reading at desks, broken up by the teacher calling on students randomly (cold-calling).	Students like JV (Under-stimulated/Hyperactive): Sitting still causes them to check out, look at their phones, or disrupt neighbors because their focus is unchanneled.	

Grade 10 English

UDL SCANNING DIMENSION	THE MONOCULTURE (CURRENT DESIGN)	PREDICTED BARRIERS	THE UDL TOOL SOLUTION
Scan the Representation	The teacher plays a 20-minute audio recording of British actors reading Macbeth in Old English while students follow along in a dense printed textbook.	Students like Elena (ELL/Visual learners): They get completely lost in the rapid British accents and archaic words. They disconnect due to language overwhelm.	The Solution: Provide a visual Graphic Novel/Komiks version of the scene summary, or use code-switching (Taglish annotations) on the board to clarify complex metaphors.
Scan the Action & Expression	The summative performance task requires every student to individually write a formal 3-page character analysis essay in English	Students like Marcus (Reluctant writers/Dysgraphia): They know who Macbeth is and understand his ambition perfectly, but the physical act of essay layout blocks them, resulting in blank papers.	
Scan the Engagement	The classroom dynamic is entirely passive: silent, individual reading at desks, broken up by the teacher calling on students randomly (cold-calling).	Students like JV (Under-stimulated/Hyperactive): Sitting still causes them to check out, look at their phones, or disrupt neighbors because their focus is unchanneled.	

Grade 10 English

UDL SCANNING DIMENSION

THE MONOCULTURE (CURRENT DESIGN)

PREDICTED BARRIERS

THE UDL TOOL SOLUTION

Scan the Representation

The teacher plays a 20-minute audio recording of British actors reading Macbeth in Old English while students follow along in a dense printed textbook.

Students like Elena (ELL/Visual learners): They get completely lost in the rapid British accents and archaic words. They disconnect due to language overwhelm.

The Solution: Provide a visual Graphic Novel/Komiks version of the scene summary, or use code-switching (Taglish annotations) on the board to clarify complex metaphors.

Scan the Action & Expression

The summative performance task requires every student to individually write a formal 3-page character analysis essay in English

Students like Marcus (Reluctant writers/Dysgraphia): They know who Macbeth is and understand his ambition perfectly, but the physical act of essay layout blocks them, resulting in blank papers.

The Solution (Choice Board): Let students choose their output: Write the essay, OR record a 3-minute podcast analyzing Macbeth like a modern political scandal, OR illustrate a storyboard tracking his downfall.

Scan the Engagement

The classroom dynamic is entirely passive: silent, individual reading at desks, broken up by the teacher calling on students randomly (cold-calling).

Students like JV (Under-stimulated/Hyperactive): Sitting still causes them to check out, look at their phones, or disrupt neighbors because their focus is unchanneled.

Grade 10 English

UDL SCANNING DIMENSION

THE MONOCULTURE (CURRENT DESIGN)

PREDICTED BARRIERS

THE UDL TOOL SOLUTION

Scan the Representation

The teacher plays a 20-minute audio recording of British actors reading Macbeth in Old English while students follow along in a dense printed textbook.

Students like Elena (ELL/Visual learners): They get completely lost in the rapid British accents and archaic words. They disconnect due to language overwhelm.

The Solution: Provide a visual Graphic Novel/Komiks version of the scene summary, or use code-switching (Taglish annotations) on the board to clarify complex metaphors.

Scan the Action & Expression

The summative performance task requires every student to individually write a formal 3-page character analysis essay in English

Students like Marcus (Reluctant writers/Dysgraphia): They know who Macbeth is and understand his ambition perfectly, but the physical act of essay layout blocks them, resulting in blank papers.

The Solution (Choice Board): Let students choose their output: Write the essay, OR record a 3-minute podcast analyzing Macbeth like a modern political scandal, OR illustrate a storyboard tracking his downfall.

Scan the Engagement

The classroom dynamic is entirely passive: silent, individual reading at desks, broken up by the teacher calling on students randomly (cold-calling).

Students like JV (Under-stimulated/Hyperactive): Sitting still causes them to check out, look at their phones, or disrupt neighbors because their focus is unchanneled.

The Solution: Run a "Pinoy Quiz-Bee" style team game where rows compete to decode a character's motives, or allow them to work in collaborative barkadas (peer groups) to analyze the text.



"The Chaos in Section Bonifacio"

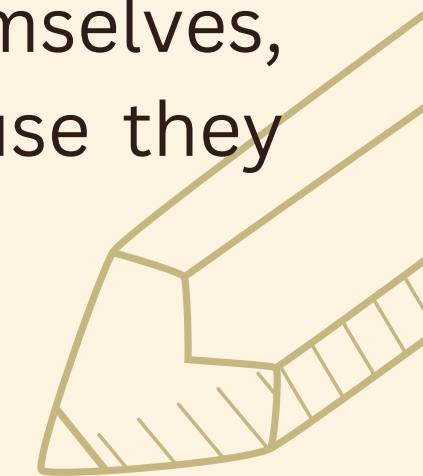
Grade 9 Science (Biology) - Ecosystems & Biodiversity

Teacher Randy's Grade 9 Science Class (Section Bonifacio)

The Goal: Explain how changes in a local ecosystem affect biodiversity and stability.

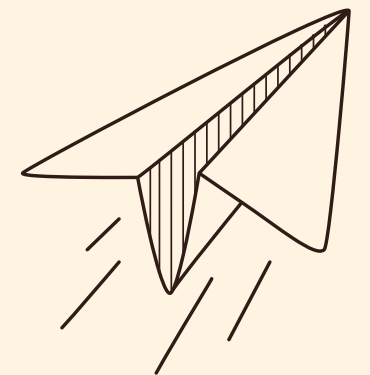
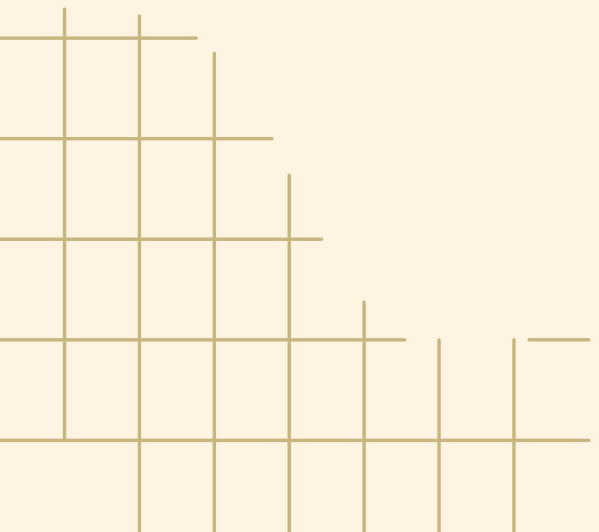
The Traditional Lesson: Teacher Randy writes long definitions of "ecosystem stability," "limiting factors," and "food webs" on the blackboard. He lectures in formal English for 35 minutes. He then commands the class to copy the notes verbatim into their science notebooks for notebook checking. For their summative assessment, students must write a 2-page, individual essay in English analyzing the ecological disruption of a local ecosystem (like the deforestation of the Sierra Madre or the pollution of Manila Bay).

The Result: Out of 45 students in the hot, crowded room, half are asleep or fanning themselves, a group at the back is making noise, and 15 students submitted blank papers because they struggled to write the essay in English.s to be active.





Core Principles of Effective Instructional Design





**"Think about the best teacher you've ever had.
What did that teacher do that helped you learn effectively?"**

17 Principles of Effective Instruction

- **Begin a lesson with a short review of previous learning.**
- **Present new material in small steps with student practice after each step.**
- **Limit the amount of material students receive at one time.**
- **Give clear and detailed instructions and explanations.**
- **Ask a large number of questions and check for understanding.**
- **Provide a high level of active practice for all students.**
- **Guide students as they begin to practice.**
- **Think aloud and model steps.**
- **Provide models of worked-out problems.**
- **Ask students to explain what they have learned.**
- **Check the responses of all students.**
- **Provide systematic feedback and corrections.**

REFLECTION QUESTIONS



- **Which principle do you already use consistently?**
- **Which principle do you find most challenging? Why?**
- **Which principle do you think would have the greatest impact on your students?**
- **If you could improve just one aspect of your teaching this term, what would it be?**
- **Which principle do you think is most often overlooked in classrooms?**

The Situation



Teacher begins her lesson by saying:

"Good morning, class! Today we'll learn about photosynthesis, cellular respiration, chlorophyll, ATP production, the Calvin Cycle, and the light-dependent reactions. These concepts are all interconnected, so please pay close attention."

Without reviewing the previous lesson on plant cells and the functions of chloroplasts, she immediately projects a complex diagram containing unfamiliar labels and begins explaining every stage of photosynthesis.

After a 20-minute lecture, she says,

"Now answer the worksheet by yourselves. If you have questions, just ask."
Several students stare silently at the worksheet, while others begin copying answers from classmates.

Your Task

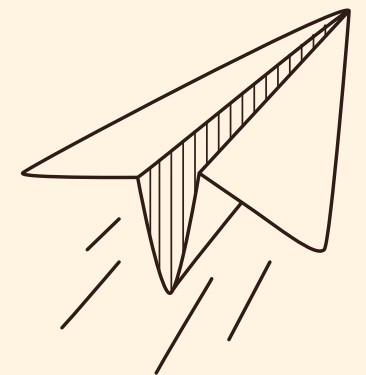


Working in groups, discuss the following:

1. What instructional errors did the teacher make during the pre-teaching phase?
2. Which of the core principles were not applied?
3. How would you revise the beginning of this lesson to better prepare students for learning?



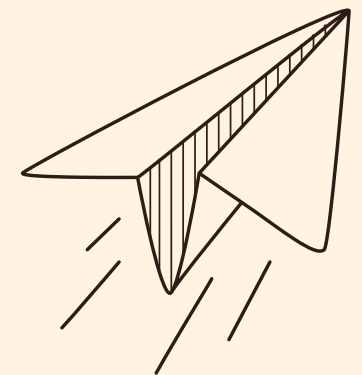
Robert Gagne's Nine Events of Instruction

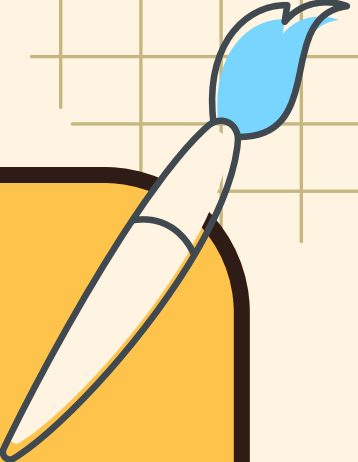
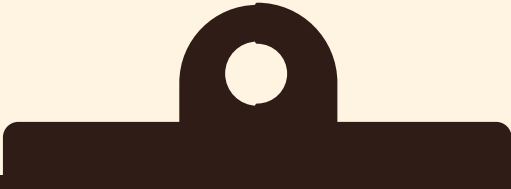




Robert Mills Gagné

Robert Gagne was an American education psychologist who focused his research in explaining and simplifying what he and others considered '**good instruction.**' Gagne is most well-known for the systematic principles he expounded within his “conditions of learning.”





Nine Events of Instruction

It is a widely used educational framework developed by Robert Gagné that outlines the optimal conditions for effective learning. It takes learners through a natural sequence, from gaining their attention to applying new knowledge in real-world scenarios to ensure thorough comprehension and skill retention



Robert Gagne's Nine Events of Instruction

1. Gain Attention

Ensure the learners are ready to learn and participate in activities by presenting a stimulus to capture their attention.

2. Inform Objectives

Inform students of the objectives or outcomes for the course and individual lessons

3. Stimulate Recall

Help students make sense of new information by relating it to something they already know or something they have already experienced.

4. Present Material

Use strategies to present and cue lesson content to provide more effective instruction. Organize and group content in meaningful ways, and provide explanations after demonstrations.

5. Guide Learning

Advise students of strategies to aid them in learning content and of resources available.

6. Elicit Performance

Have students apply what they have learned to reinforce new skills and knowledge and to confirm correct understanding of course concepts.

7. Provide Feedback

Provide timely feedback of students' performance to assess and facilitate learning and to allow students to identify gaps in understanding before it is too late.

8. Assess performance

Test whether the expected learning outcomes have been achieved on previously stated course objectives.

9. Enhance Retention and Transfer

Help learners retain more information by providing them opportunities to connect course concepts to potential real-world applications.



The Lesson Planning Blueprint

Phase	Event	Classroom Term	What the Teacher Does	Why it Matters
Phase 1 : Set up	Gain Attention	The Hook	Use a demo, mystery object, or provocative question	Alerts the brain; focuses wandering minds
	Inform Objectives	The Goal	Write “I can” statement clearly on board	Gives students a target; builds motivation
	Stimulate Recall	The Bridge	Do a quick 2-minute review of yesterday’s lesson	Gives students a target; builds motivation
Phase 2: Delivery	Present Material	I Do	Teach the core content in small, chunked, 10-minute bursts.	Avoids cognitive overload; keeps kids focused.
	Guide Learning	We Do	Use graphic organizers, analogies, or step-by-step modeling.	Helps the brain store data correctly.
	Elicit Performance	You Do	Have students try it via think-pair-share, whiteboards, or worksheets.	Forces students to actively retrieve information.

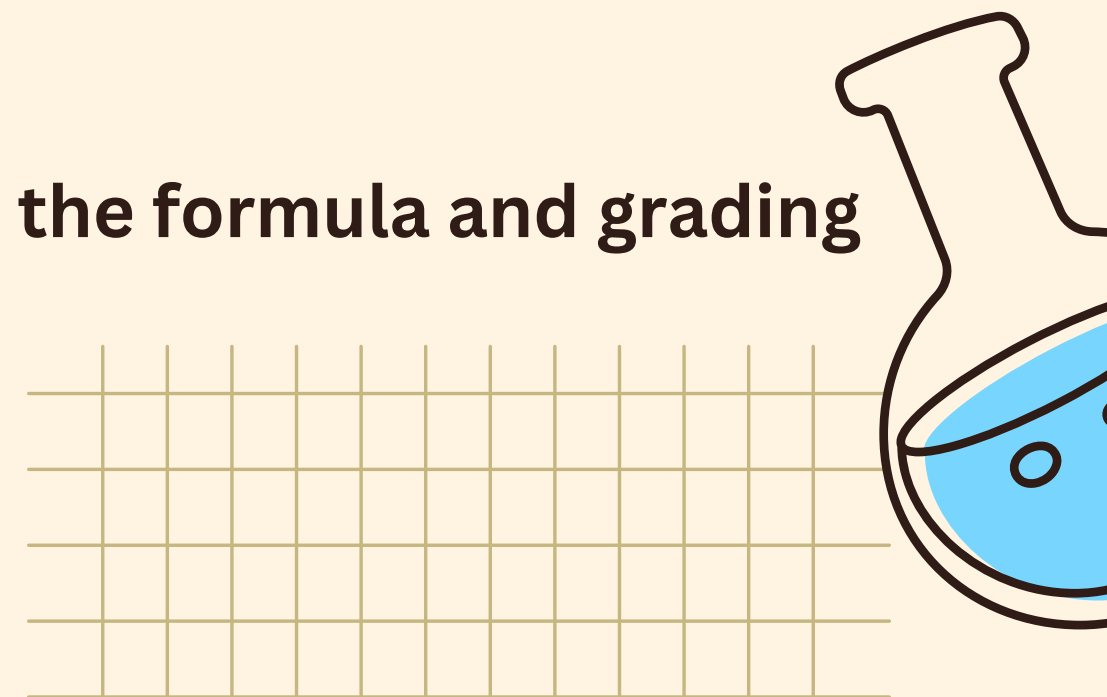
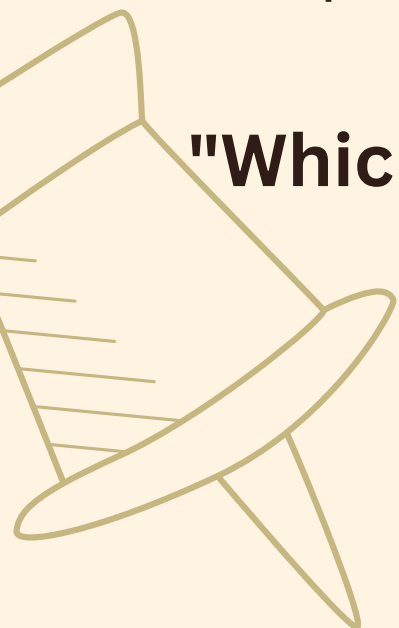
The Lesson Planning Blueprint

Phase	Event	Classroom Term	What the Teacher Does	Why it Matters
Phase 3: Lock-In	Provide Feedback	Check for Understanding	Walk around the room; give immediate, corrective pointers.	Prevents students from practicing mistakes.
	Assess Performance	Exit Ticket / Quiz	Administer a short, independent check without teacher help.	Proves who mastered the concept today.
	Enhance Transfer	Real-World Connection	Assign a creative homework task or connect to a future project.	Answers the student question: "Why do we need this?"

Let's Improve

Mr. Davis begins his fifth-grade math lesson on calculating the area of a triangle by showing a picture of the Great Pyramid of Giza and asking his students how much stone it took to cover the sides, successfully gaining their attention. He then informs them of the lesson objectives for by writing on the board that they will learn how to find the area of any triangle using a formula. To stimulate recall, he quickly reminds the class that they learned how to find the area of a rectangle using length times width the previous week. Next, he presents the new material for by introducing the triangle area formula and writing out two step-by-step examples on the whiteboard. He hands out a worksheet with ten triangles and instructs the students to spend fifteen minutes solving them on their own. As they work, he provides formative feedback for by walking around the room to offer immediate corrections and praise. Finally, to assess performance, he has the students independently complete a two-question Exit Ticket at the end of class.

"Which crucial instructional event did Mr. Davis skip between presenting the formula and grading their worksheets?"



FEEDBACK



2030

Thank You

Thank you for your attention and dedication in supporting joint efforts to improve the quality of education.



